# $\SHEAFHOM$ AND $\MATHIT{ISOM}$ OF LOCALLY CONSTANT SHEAVES OF MODULES ARE LOCALLY CONSTANT

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*Editorial note.* All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate and re-verified against the pinned source (sm-twins-20261014.json span sanity: all OK).

*Editorial note.* Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found hard occupation only from the 47 pre-existing lane-s11 S11-sites-mod-* cards (same pinned snapshot, delivered) and cross-chapter citation noise, 0 other-lane hits for sites-modules.tex (sm-occupied.json, 2026-10-14); the RESUME-pool dispatch unlocks the remaining face.

*Source reference.* Original lemma 093N: sources/stacks-chow/tags.txt, source lines 11521–11521. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

**Lemma 0.1.** *Let $\mathcal{C}$ be a site. Let $\Lambda$ be a ring. Let $M$, $N$ be $\Lambda$-modules. Let $\mathcal{F}, \mathcal{G}$ be a locally constant sheaves of $\Lambda$-modules.*

(1) *If $M$ is of finite presentation, then*
$$\underline{\mathrm{Hom}_\Lambda(M, N)} = \mathcal{H}om_{\underline{\Lambda}}(\underline{M}, \underline{N})$$
(2) *If $M$ and $N$ are both of finite presentation, then*
$$\underline{Isom_\Lambda(M, N)} = Isom_{\underline{\Lambda}}(\underline{M}, \underline{N})$$
(3) *If $\mathcal{F}$ is of finite presentation, then $\mathcal{H}om_{\underline{\Lambda}}(\mathcal{F}, \mathcal{G})$ is a locally constant sheaf of $\Lambda$-modules.*
(4) *If $\mathcal{F}$ and $\mathcal{G}$ are both of finite presentation, then $Isom_{\underline{\Lambda}}(\mathcal{F}, \mathcal{G})$ is a locally constant sheaf of sets.*

*Proof.* Proof of (1). Set $E = \mathrm{Hom}_\Lambda(M, N)$. We want to show the canonical map
$$\underline{E} \longrightarrow \mathcal{H}om_{\underline{\Lambda}}(\underline{M}, \underline{N})$$
is an isomorphism. The module $M$ has a presentation $\Lambda^{\oplus s} \to \Lambda^{\oplus t} \to M \to 0$. Then $E$ sits in an exact sequence
$$0 \to E \to \mathrm{Hom}_\Lambda(\Lambda^{\oplus t}, N) \to \mathrm{Hom}_\Lambda(\Lambda^{\oplus s}, N)$$
and we have similarly
$$0 \to \mathcal{H}om_{\underline{\Lambda}}(\underline{M}, \underline{N}) \to \mathcal{H}om_{\underline{\Lambda}}(\underline{\Lambda^{\oplus t}}, \underline{N}) \to \mathcal{H}om_{\underline{\Lambda}}(\underline{\Lambda^{\oplus s}}, \underline{N})$$

*Date*: stacks-project:sites-modules.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

This reduces the question to the case where $M$ is a finite free module where the result is clear.

Proof of (3). The question is local on $\mathcal{C}$, hence we may assume $\mathcal{F} = \underline{M}$ and $\mathcal{G} = \underline{N}$ for some $\Lambda$-modules $M$ and $N$. By Lemma 093N the module $M$ is of finite presentation. Thus the result follows from (1).

Parts (2) and (4) follow from parts (1) and (3) and the fact that $Isom$ can be viewed as the subsheaf of sections of $\mathcal{H}om_{\underline{\Lambda}}(\mathcal{F}, \mathcal{G})$ which have an inverse in $\mathcal{H}om_{\underline{\Lambda}}(\mathcal{G}, \mathcal{F})$. $\square$